

BRETT H. ANDREWS

Curriculum Vitae

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EXPERIENCE	Research Associate Professor	2022–present
	Department of Physics and Astronomy, University of Pittsburgh	
	Research Assistant Professor	2017–2022
	Department of Physics and Astronomy, University of Pittsburgh	
	Postdoctoral Associate	2014–2017
	Department of Physics and Astronomy, University of Pittsburgh	

EDUCATION	Ph.D. Astronomy	2014
	The Ohio State University “Decoding Galaxy Evolution with Gas-phase and Stellar Elemental Abundances” Advisors: David H. Weinberg & Jennifer A. Johnson	
	M.S. Astronomy	2011
	The Ohio State University	
	B.S. Physics & Astronomy	2008
	Yale University Nicolas Adamo Scholar-Athlete Prize (Silliman College)	

PUBLICATIONS	4 lead author; 13 significant contributing author; 61 contributing author; 3 additional publications. Over 17,000 citations. List attached.
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PRESENTATIONS	5 invited; 31 contributed; 2 posters. List attached.
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FUNDING	Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities	2025–present
	Exploiting the Intrinsic Correlations Between Galaxy Properties: Redshift and Beyond \$299,887	
	PI: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities	2023–present
	Exploiting Deep Learning to Improve Roman Photometric Redshifts \$287,881	
	Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities	2023–present
	A Statistical Framework for Optimizing Roman Spectroscopic Training Sets \$291,061	

	Co-I: NASA Nancy Grace Roman Space Telescope Research and Support Participation Opportunities Maximizing Cosmological Science with the Roman High Latitude Imaging Survey \$19.5M (\$665,310 to Pitt)	2023–present
	Co-I: NASA Hubble Space Telescope Cycle 30 SNAP Proposal (409 orbits) Post-starbursts from DESI: Timing quenching and morphological transformation at $1 < z < 1.3$ \$202,893	2023–2025
	Senior Personnel: NSF Major Research Instrumentation Program Acquisition of Cutting-Edge GPU and MPI Nodes for the Interdisciplinary Pitt Center for Research Computing \$1,187,606	2021–2024
	PI: NSF Astronomy and Astrophysics Research Grants Interpretable and Deblended Photometric Redshifts with a Deep Capsule Network \$535,578	2020–2025
	Co-I: NASA Astrophysical Data Analysis Program Multiwavelength Milky Way Analogs \$504,949	2019–2024
	Co-I: NASA Astrophysical Data Analysis Program (Funded Extension) Multiwavelength Milky Way Analogs \$33,707	2021
MENTORING PH.D.	NASA PA Space Grant Consortium Research Scholarship Awards In support of undergraduate research: Zach Lewis: Fall 2019, Fall 2020, Spring 2021, Fall 2021, Spring 2022 Katie Mack: Summer 2021, Fall 2021, Spring 2022 Emma Moran: Spring 2024, Summer 2024, Fall 2024, Spring 2025 Total: \$32,500	2019–present
	Marcos Tamargo-Arizmendi (University of Pittsburgh) “Modeling Incompleteness in Spectroscopic Redshift Datasets”	2025–present
	Finian Ashmead (University of Pittsburgh) “A Statistical Framework for Optimizing Roman Spectroscopic Training Sets for Roman Photometric Redshifts”	2023–present
	Yoquelbin Salcedo Hernandez (University of Pittsburgh) “DESI-2 Emission Line Galaxy Target Selection”	2022–present
	Ashod Khederlarian (University of Pittsburgh) “Towards Leveraging Deep Learning for Photometric Redshifts with the Roman Space Telescope”	2022–present
	Biprateep Dey (University of Pittsburgh) “Cosmic Cartography: Photometric Redshifts for Next-Generation Sky Surveys” <i>Next Position:</i> Schmidt AI for Science Fellowship at University of Toronto + CITA Fellowship + Dunlap Institute Fellowship	2018–2024

	Troy Raen (University of Pittsburgh) “Toward the Study of Stars with LSST” <i>Next Position:</i> Application Developer at IPAC/Caltech	2019–2022
	Catherine Fielder (University of Pittsburgh) “Constraining the Milky Way’s Ultraviolet to Infrared SED” <i>Next Position:</i> Postdoctoral Researcher at University of Arizona	2018–2022
	Ph.D. Thesis Committees (University of Pittsburgh) – Julissa Sarmiento	2025–present
M.S.	Quanbin (Eric) Ma (Carnegie Mellon University Machine Learning Department) “Deep Dimensionality Reduction of MaNGA Data” <i>Next Position:</i> Software Engineer at Facebook	2017
Undergraduate	Emma Moran (University of Pittsburgh) “Deep Learning Improves Photometric Redshifts in All Regions of Color Space”	2023–2025
	Zach Lewis (University of Pittsburgh) “The Gas-Phase Mass-Metallicity Relation for Massive Galaxies at $z \sim 0.7$ with the LEGA-C Survey” <i>Next Position:</i> NSF GRFP Fellow and Astronomy Ph.D. Student at the University of Wisconsin	2019–2022
	Katie Mack (University of Pittsburgh) “Comparing Stellar and Emission Line Dust Attenuation in the LEGA-C Survey”	2021–2022
	Ian Cooper (University of Pittsburgh) “SNIa Hosts in MaNGA” <i>Next Position:</i> Geophysics Ph.D. Student at Boston College	2016
	Mariah Jones (Baldwin High School) “The Mass–Metallicity Relation at Intermediate Redshifts” <i>Next Position:</i> Undergraduate Student at Vassar College	2020–2021
SERVICE TO PROFESSION	Invited Reviewer – <i>NSF</i> – <i>NASA</i> – <i>Swiss NSF</i>	
	Invited Referee – <i>The Astrophysical Journal</i> – <i>Monthly Notices of the Royal Astronomical Society</i> – <i>Astronomy & Astrophysics</i> – <i>Publications of the Astronomical Society of the Pacific</i>	
	Faculty Hiring Committee Department of Physics and Astronomy, University of Pittsburgh – LSST LINCC Research Assistant Professor – Astrophysics or Experimental High Energy Physics with Machine Learning (Tenure-stream)	2022–2023 2021–2022 2018–2019

	Research Faculty Promotion Criteria Committee University of Pittsburgh Dietrich School of Arts and Sciences	2024
	Union of Pitt Faculty (USW Local 1088-04) University of Pittsburgh – Research Stream Advocacy Committee (Founding Chair) – Bargaining Committee – Organizing Committee – Steward	2025–present 2025–present 2024–present 2024–present
	Faculty Peer Buddy University of Pittsburgh	2021–2023
	Faculty Mentor LSST Dark Energy Science Collaboration (DESC)	2025
	Astrophysics Academic Program Committee University of Pittsburgh	2025
TEACHING	ASTRON 3580: Galactic and Extragalactic Astronomy (University of Pittsburgh) Co-instructor with Jeff Newman.	Spring 2024
	Member of the Graduate Faculty (University of Pittsburgh)	2024–present
WORKSHOPS	AstroPGH Python Boot Camp and Summer Seminar Series (University of Pittsburgh and Carnegie Mellon University) • Intense three day workshop introducing the Python programming language for astrophysics/physics graduate and undergraduate students. – Organizer – Organizer. Presented lecture on Matplotlib. – Organizer – Organizer. Presented lecture on Python and Jupyter. – Organizer. Presented two lectures on Numpy and Pandas. – Organizer. Presented eight lectures on Python basics, Astropy, Pandas, Git, and GitHub.	2025 2024 2023 2022 2021 2020
	Writing Fridays Workshops (University of Pittsburgh) • Led weekly writing workshops for astrophysics graduate students.	2023–present
	Professional Development Workshops (University of Pittsburgh) • Co-led workshops with Rachel Bezanson for astrophysics and physics graduate students. – CV, Resume, and Webpage Workshop – Webpage and CV Workshops – Webpage Workshop – CV, Resume, and Webpage Workshops	2024 2022 2021 2020
	Data Science Group Meeting (University of Pittsburgh) • Organized event series for astrophysics graduate students.	2018
	Marvin Workshops (SDSS-IV Collaboration Meetings) • Led introductory workshops for users of the Marvin software toolkit.	2015–2017

	Python Boot Camps (University of Pittsburgh) • Led events for undergraduate and graduate students.	2015
SCIENCE COMMUNICATION	The Ellis School (First Grade STEM Class) – “How the Sun, Earth, and Moon align for a solar eclipse” Shadyside Presbyterian Church Nursery School (Pre-K Class) – “A Few Beautiful Minutes: Experiencing a Solar Eclipse” Allegheny Observatory Public Lecture Series – “The True Colors of the Milky Way” Astronomy on Tap – “The Origin of the Elements” – “Indiana Jones and the Hidden Galaxy” w/ Courtney Epstein Learn & Earn Corporate Host Hosted intern for summer youth employment program delivered by Allegheny County, the City of Pittsburgh, and Partner4Work. Science Olympiad Coached Grandview Heights (Ohio) Middle School team.	2024 2024 2023 2018 2014 2021 2011
COLLAB.	Roman Science Collaboration Member Roman High Latitude Imaging Survey (HLIS) Cosmology Project Infrastructure Team (PIT) Member Dark Energy Spectroscopic Instrument (DESI) Member LSST Dark Energy Science Collaboration (DESC) Full Member SDSS-IV/MaNGA Architect	2025– 2024– 2024– 2023– 2016
SOFTWARE	flexCE Python package for modeling galactic chemical evolution. Marvin Python package, RESTful API, and Flask web application for accessing, visualizing, and analyzing SDSS-IV MaNGA integral-field spectroscopic data.	

BRETT H. ANDREWS

Publications

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|-------------------------|---|
| LEAD AUTHOR | <ol style="list-style-type: none"> 4. Andrews, B. H., et al., 2017, <i>The Astrophysical Journal</i>, 835, 224
Inflow, Outflow, Yields, and Stellar Population Mixing in Chemical Evolution Models 3. Andrews, B. H. & Martini, P., 2013, <i>The Astrophysical Journal</i>, 765, 140
The Mass-Metallicity Relation with the Direct Method on Stacked Spectra of SDSS Galaxies 2. Andrews, B. H., et al., 2012, <i>Acta Astronomica</i>, 62, 269
Principal Component Abundance Analysis of Microlensed Bulge Dwarf and Subgiant Stars 1. Andrews, B. H. & Thompson, T. A., 2011, <i>The Astrophysical Journal</i>, 727, 97
Assessing Radiation Pressure as a Feedback Mechanism in Star-forming Galaxies |
| SIG. CONTRIB.
AUTHOR | <ol style="list-style-type: none"> 13. Ashmead, F., Newman, J. A., Andrews, B. H., et al., 2025, arXiv:2512.09032, submitted to <i>The Astrophysical Journal</i>
Optimizing Photometric Redshift Training Sets I: Efficient Compression of the Galaxy Color-Redshift Relation with UMAP 12. Moran, E. R., Andrews, B. H., Newman, J. A., et al., 2025, arXiv:2507.06299, accepted to <i>The Astrophysical Journal</i>
Deep Learning Improves Photometric Redshifts in All Regions of Color Space 11. Dey, B., Zhao, D., Andrews, B. H., et al., 2025, <i>Machine Learning: Science and Technology</i>, 6, 045058
Towards instance-wise calibration: local amortized diagnostics and reshaping of conditional densities (LADaR) 10. Khederlarian, A., Newman, J. A., Andrews, B. H., et al., 2024, <i>Monthly Notices of the Royal Astronomical Society</i>, 531, 1454
Emission Line Predictions for Mock Galaxy Catalogues: a New Differentiable and Empirical Mapping from DESI 9. Lewis, Z., Andrews, B. H., et al., 2024, <i>The Astrophysical Journal</i>, 964, 59
The Gas-Phase Mass–Metallicity Relation for Massive Galaxies at $z \sim 0.7$ with the LEGA-C Survey 8. Fielder, C., Andrews, B. H., et al., 2023, <i>Monthly Notices of the Royal Astronomical Society</i>, 525, 1023
Empirically Driven Multiwavelength K-corrections At Low Redshift 7. Kodra, D., Andrews, B. H., et al., 2023, <i>The Astrophysical Journal</i>, 942, 36
Optimized Photometric Redshifts for the Cosmic Assembly Near-Infrared Deep Extragalactic Legacy Survey (CANDELS) 6. Dey, B., Andrews, B. H., et al., 2022, <i>Monthly Notices of the Royal Astronomical Society</i>, 515, 4
Photometric Redshifts from SDSS Images with an Interpretable Deep Capsule Network 5. Dey, B., Newman, J. A., Andrews, B. H., et al., 2021, arXiv:2110.15209
Re-calibrating Photometric Redshift Probability Distributions Using Feature-space Regression 4. Fielder, C. E., Newman, J. A., Andrews, B. H., et al., 2021, <i>Monthly Notices of the Royal Astronomical Society</i>, 508, 4459
Constraining the Milky Way’s ultraviolet-to-infrared SED with Gaussian process regression 3. Cherinka, B., Andrews, B. H., et al., 2019, <i>The Astronomical Journal</i>, 158, 74
Marvin: A Tool Kit for Streamlined Access and Visualization of the SDSS-IV MaNGA Data Set 2. Weinberg, D. H., Andrews, B. H., & Freudenburg, J., 2017, <i>The Astrophysical Journal</i>, 837, 183
Equilibrium and Sudden Events in Chemical Evolution |

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|------------------------|--|
| | <p>1. Brown, J. S., Martini, P., & Andrews, B. H., 2016, <i>Monthly Notices of the Royal Astronomical Society</i>, 458, 1529
A recalibration of strong-line oxygen abundance diagnostics via the direct method and implications for the high-redshift universe</p> |
| CONTRIBUTING
AUTHOR | <p>61. Zhang, T., et al., 2025, arXiv:2510.07370, submitted to <i>Monthly Notices of the Royal Astronomical Society</i>
Photometric Redshift Estimation for Rubin Observatory Data Preview 1 with Redshift Assessment Infrastructure Layers (RAIL)</p> <p>60. Zhang, T., et al., 2025, arXiv:2507.01374, submitted to <i>Monthly Notices of the Royal Astronomical Society</i>
Forecasting the Impact of Source Galaxy Photometric Redshift Uncertainties on the LSST 3×2pt Analysis</p> <p>59. Zhang, Y., et al., 2024, <i>The Astrophysical Journal</i>, 976, 36
DESI Massive Post-Starburst Galaxies at $z \sim 1.2$ have compact structures and dense cores</p> <p>58. Moskowit, I., et al., 2024, <i>The Astrophysical Journal Letters</i>, 967, L6
Improving Photometric Redshift Estimates with Training Sample Augmentation</p> <p>57. Yantovski-Barth, M. J., et al., 2024, <i>Monthly Notices of the Royal Astronomical Society</i>, 531, 2285
The CluMPR Galaxy Cluster-Finding Algorithm and DESI Legacy Survey Galaxy Cluster Catalogue</p> <p>56. Zhou, S., et al., 2023, <i>Monthly Notices of the Royal Astronomical Society</i>, 521, 5810
Are Milky-Way-like galaxies like the Milky Way? A view from SDSS-IV/MaNGA</p> <p>55. Setton, D., et al., 2023, <i>The Astrophysical Journal Letters</i>, 947, 31
DESI Survey Validation Spectra Reveal an Increasing Fraction of Recently Quenched Galaxies at $z \sim 1$</p> <p>54. Kartaltepe, J., et al., 2023, <i>The Astrophysical Journal Letters</i>, 946, 15
CEERS Key Paper. III. The Diversity of Galaxy Structure and Morphology at $z = 3-9$ with JWST</p> <p>53. Boardman, N., et al., 2022, <i>Monthly Notices of the Royal Astronomical Society</i>, 514, 2298
How well do local relations predict gas-phase metallicity gradients? Results from SDSS-IV MaNGA</p> <p>52. Oyarzún, G. A., et al., 2022, <i>The Astrophysical Journal</i>, 933, 88
SDSS-IV MaNGA: How the Stellar Populations of Passive Central Galaxies Depend on Stellar and Halo Mass</p> <p>51. Schaefer, A. L., et al., 2022, <i>The Astrophysical Journal</i>, 930, 2
SDSS-IV MaNGA: Exploring the Local Scaling Relations for N/O</p> <p>50. Abdurro'uf, et al., 2021, <i>The Astrophysical Journal Supplement Series</i>, 259, 2
The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar, and APOGEE-2 Data</p> <p>49. Law, D., et al., 2020, <i>The Astrophysical Journal</i>, 915, 35
SDSS-IV MaNGA: Refining Strong Line Diagnostic Classifications Using Spatially Resolved Gas Dynamics</p> <p>48. Parikh, T., et al., 2021, <i>Monthly Notices of the Royal Astronomical Society</i>, 502, 5508
SDSS-IV MaNGA: radial gradients in stellar population properties of early-type and late-type galaxies</p> <p>47. Greener, M., et al., 2021, <i>Monthly Notices of the Royal Astronomical Society</i>, 502, 95
SDSS-IV MaNGA: the 'G-dwarf problem' revisited</p> <p>46. Luo, Y., et al., 2021, <i>The Astrophysical Journal</i>, 908, 183
Evidence for the Accretion of Gas in Star-forming Galaxies: High N/O Abundances in Regions of Anomalously Low Metallicity</p> <p>45. Mazzola, C., et al., 2020, <i>Monthly Notices of the Royal Astronomical Society</i>, 499, 1607
The close binary fraction as a function of stellar parameters in APOGEE: a strong anticorrelation with α abundances</p> |

44. Boardman, N., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 498, 4943
Are the Milky Way and Andromeda unusual? A comparison with Milky Way and Andromeda analogues
43. Fraser-McKelvie, A., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 495, 4158
SDSS-IV MaNGA: spatially resolved star formation in barred galaxies
42. Ahumada, R., et al., 2020, *The Astrophysical Journal Supplement Series*, 249, 3
The 16th Data Release of the Sloan Digital Sky Surveys: First Release from the APOGEE-2 Southern Survey and Full Release of eBOSS Spectra
41. Greener, M. J., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 495, 2305
SDSS-IV MaNGA: spatially resolved dust attenuation in spiral galaxies
40. Schaefer, A. L., et al., 2020, *The Astrophysical Journal*, 890, L3
SDSS-IV MaNGA: Variations in the N/O-O/H Relation Bias Metallicity Gradient Measurements
39. Boardman, N., et al., 2020, *Monthly Notices of the Royal Astronomical Society*, 491, 3672
Milky Way analogues in MaNGA: multiparameter homogeneity and comparison to the Milky Way
38. Westfall, K. B., et al., 2019, *The Astronomical Journal*, 158, 231
The Data Analysis Pipeline for the SDSS-IV MaNGA IFU Galaxy Survey: Overview
37. Fraser-McKelvie, A., et al., 2019, *Monthly Notices of the Royal Astronomical Society*, 488, L6
SDSS-IV MaNGA: stellar population gradients within barred galaxies
36. Pace, Z. J., et al., 2019, *The Astrophysical Journal*, 883, 83
Resolved and Integrated Stellar Masses in the SDSS-IV/MaNGA Survey. II. Applications of PCA-based Stellar Mass Estimates
35. Pace, Z. J., et al., 2019, *The Astrophysical Journal*, 883, 82
Resolved and Integrated Stellar Masses in the SDSS-IV/MaNGA Survey. I. PCA Spectral Fitting and Stellar Mass-to-light Ratio Estimates
34. Zhang, K., et al., 2019, *The Astrophysical Journal*, 883, 63
Machine-learning Classifiers for Intermediate Redshift Emission-line Galaxies
33. Oyarzún, G. A., et al., 2019, *The Astrophysical Journal*, 880, 111
Signatures of Stellar Accretion in MaNGA Early-type Galaxies
32. Parikh, T., et al., 2019, *Monthly Notices of the Royal Astronomical Society*, 483, 3420
SDSS-IV MaNGA: local and global chemical abundance patterns in early-type galaxies
31. Aguado, D. S., et al., 2019, *The Astrophysical Journal Supplement Series*, 240, 23
The Fifteenth Data Release of the Sloan Digital Sky Surveys: First Release of MaNGA-derived Quantities, Data Visualization Tools, and Stellar Library
30. Hwang, H.-C., et al., 2019, *The Astrophysical Journal*, 872, 144
Anomalously Low-metallicity Regions in MaNGA Star-forming Galaxies: Accretion Caught in Action?
29. Li, H., et al., 2019, *The Astrophysical Journal*, 872, 63
Interpreting the Star Formation-Extinction Relation with MaNGA
28. Zasowski, G., et al., 2019, *The Astrophysical Journal*, 870, 138
APOGEE DR14/DR15 Abundances in the Inner Milky Way
27. Rowlands, K., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 480, 2544
SDSS-IV MaNGA: spatially resolved star formation histories and the connection to galaxy physical properties
26. Parikh, T., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 477, 3954
SDSS-IV MaNGA: the spatially resolved stellar initial mass function in ~ 400 early-type galaxies
25. Penny, S. J., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 476, 979
SDSS-IV MaNGA: evidence of the importance of AGN feedback in low-mass galaxies
24. Abolfathi, B., et al., 2018, *The Astrophysical Journal Supplement Series*, 235, 42
The Fourteenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the Extended Baryon Oscillation Spectroscopic Survey and from the Second Phase of the Apache Point Observatory Galactic Evolution Experiment

23. Talbot, M. S., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 477, 195
SDSS-IV MaNGA: the spectroscopic discovery of strongly lensed galaxies
22. Wylezalek, D., et al., 2018, *Monthly Notices of the Royal Astronomical Society*, 474, 1499
SDSS-IV MaNGA: identification of active galactic nuclei in optical integral field unit surveys
21. Badenes, C., et al., 2018, *The Astrophysical Journal*, 854, 147
Stellar Multiplicity Meets Stellar Evolution and Metallicity: The APOGEE View
20. Albareti, F. D., et al., 2017, *The Astrophysical Journal Supplement Series*, 233, 25
The 13th Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-IV Survey Mapping Nearby Galaxies at Apache Point Observatory
19. Greene, J. E., et al., 2017, *The Astrophysical Journal*, 851, L33
SDSS-IV MaNGA: Probing the Kinematic Morphology-Density Relation of Early-type Galaxies with MaNGA
18. Zasowski, G., et al., 2017, *The Astronomical Journal*, 154, 198
Target Selection for the SDSS-IV APOGEE-2 Survey
17. Zahid, H. J., et al., 2017, *The Astrophysical Journal*, 847, 18
Stellar Absorption Line Analysis of Local Star-forming Galaxies: The Relation between Stellar Mass, Metallicity, Dust Attenuation, and Star Formation Rate
16. Martínez-Rodríguez, H., et al., 2017, *The Astrophysical Journal*, 843, 35
Observational Evidence for High Neutronization in Supernova Remnants: Implications for Type Ia Supernova Progenitors
15. Blanton, M. R., et al., 2017, *The Astronomical Journal*, 154, 28
Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe
14. Linden, S. T., et al., 2017, *The Astrophysical Journal*, 842, 49
Timing the Evolution of the Galactic Disk with NGC 6791: An Open Cluster with Peculiar High- α Chemistry as Seen by APOGEE
13. Zhang, K., et al., 2017, *Monthly Notices of the Royal Astronomical Society*, 466, 3217
SDSS-IV MaNGA: the impact of diffuse ionized gas on emission-line ratios, interpretation of diagnostic diagrams and gas metallicity measurements
12. Jones, A., et al., 2017, *Astronomy and Astrophysics*, 599, A141
SDSS IV MaNGA: Deep observations of extra-planar, diffuse ionized gas around late-type galaxies from stacked IFU spectra
11. Yan, R., et al., 2016, *The Astronomical Journal*, 152, 197
SDSS-IV MaNGA IFS Galaxy Survey—Survey Design, Execution, and Initial Data Quality
10. Law, D. R., et al., 2016, *The Astronomical Journal*, 152, 83
The Data Reduction Pipeline for the SDSS-IV MaNGA IFU Galaxy Survey
9. Holtzman, J. A., et al., 2015, *The Astronomical Journal*, 150, 148
Abundances, Stellar Parameters, and Spectra from the SDSS-III/APOGEE Survey
8. Hayden, M. R., et al., 2015, *The Astrophysical Journal*, 808, 132
Chemical Cartography with APOGEE: Metallicity Distribution Functions and the Chemical Structure of the Milky Way Disk
7. Alam, S., et al., 2015, *The Astrophysical Journal Supplement Series*, 219, 12
The Eleventh and Twelfth Data Releases of the Sloan Digital Sky Survey: Final Data from SDSS-III
6. Nidever, D. L., et al., 2014, *The Astrophysical Journal*, 796, 38
Tracing Chemical Evolution over the Extent of the Milky Way’s Disk with APOGEE Red Clump Stars
5. Bovy, J., et al., 2014, *The Astrophysical Journal*, 790, 127
The APOGEE Red-clump Catalog: Precise Distances, Velocities, and High-resolution Elemental Abundances over a Large Area of the Milky Way’s Disk
4. Ahn, C. P., et al., 2014, *The Astrophysical Journal Supplement Series*, 211, 17
The Tenth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Apache Point Observatory Galactic Evolution Experiment

3. Leja, J., et al., 2013, *The Astrophysical Journal*, 778, L24
Exploring the Chemical Link between Local Ellipticals and Their High-redshift Progenitors
2. Zasowski, G., et al., 2013, *The Astronomical Journal*, 146, 81
Target Selection for the Apache Point Observatory Galactic Evolution Experiment (APOGEE)
1. Ahn, C. P., et al., 2012, *The Astrophysical Journal Supplement Series*, 203, 21
The Ninth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Baryon Oscillation Spectroscopic Survey

ADDITIONAL

3. Cherinka, B., et al., 2020, *ASPC*, 527, 743C
Marvin: A Toolkit for Streamlined Access and Visualisation of the SDSS-IV MaNGA Data Set
2. Wilson, J. C., et al., 2012, *SPIE*, 8446, 84460H
Performance of the Apache Point Observatory Galactic Evolution Experiment (APOGEE) high-resolution near-infrared multi-object fiber spectrograph
1. **Andrews, B. H.** & Thompson, T. A., 2011, *EAS Publications Series*, 52, 275
Radiation Pressure Feedback in Galaxies

BRETT H. ANDREWS

Presentations

INVITED	Optimizing Roman Photometric Redshifts for HLIS	2025
	Roman Virtual Lecture Series	
	Effective Plotting	2021
	DESI Collaboration Meeting	
	Hidden in Plain Sight: a Deep Learning Approach to Finding Supernovae in Galaxy Maps	2018
	Science 2018, University of Pittsburgh	
	The SDSS-IV MaNGA Survey: Galaxy Dissection on an Industrial Scale	2017
CONTRIBUTED	CMU Astro Seminar, Carnegie Mellon University	
	The Mass–Metallicity Relation in SDSS Using Electron Temperature Measurements	2015
	Understanding Nebular Emission in High-Redshift Galaxies, The Carnegie Observatories	
	Optimizing Roman Photometric Redshifts for HLIS	2025
	Cosmic Cartography with Roman, Space Telescope Science Institute	
	Optimizing Roman Photometric Redshifts	2024
	Roman HLIS Cosmology PIT Meeting, California Institute of Technology	
	Photometric Redshifts for Next Generation Imaging Surveys	2024
	DHWFest, University of Utah	
	Jumpstart Your Paper	2022
	AstroPGH Summer Seminar, University of Pittsburgh	
	AstroCoffee Tips	2022
	AstroPGH Summer Seminar, University of Pittsburgh	
	Leveraging Statistics and Machine Learning for Probing Galaxy Evolution and Measuring Galaxy Distances	2022
	AstroLunch Seminar, University of Pittsburgh	
	Effective Plotting	2021
	AstroPGH Summer Seminar, University of Pittsburgh	
	Effective Plotting	2020
	AstroPGH Summer Seminar, University of Pittsburgh	
	Effective Plotting	2018
	Astro Student Seminar, University of Pittsburgh	
	GitHub Flow	2018
	Astro Student Seminar, University of Pittsburgh	
	MaNGA DAP	2017
	SDSS-IV Collaboration Meeting, Pontificia Universidad Católica de Chile	
	Streamlining MaNGA Data with Marvin	2017
	AstroLunch, University of Pittsburgh	
	The State of Marvin	2016
	SDSS-IV Collaboration Meeting, University of Wisconsin	
	AstroCoffee Presentation Advice	2016
	Astro Student Seminar, University of Pittsburgh	
	Marvin-tools: Distilling DAP Measurements and & Stacking Spectra	2016
	MaNGA Collaboration Meeting (Cocoa Beach, FL)	
	Global vs. Resolved Metallicities	2015
	Astro Student Seminar, University of Pittsburgh	
	Global vs. Resolved Metallicities	2015
	SDSS-IV Collaboration Meeting, Instituto Física Teórica Universidad Autónoma de Madrid	
	MaNGA Data Analysis Pipeline Quality Assessment	2015
	MaNGA Collaboration Meeting, University of Kentucky	
	Principal Component Abundance Analysis: APOGEE and flexCE	2015
	Local Group Astrostatistics Conference, University of Michigan	

	Understanding the Bimodality in $[\alpha/\text{Fe}]$ with a Chemical Evolution Model SDSS-III Collaboration Meeting (Park City, UT)	2014
	Applying Principal Component Analysis to APOGEE SDSS-IV Collaboration Meeting (Park City, UT)	2014
	Exploiting Large Multi-element Stellar Abundance Surveys AAS Winter Meeting	2014
	Decoding Galactic Chemical Evolution with Gas-phase and Stellar Abundances Colloquium, University of Wisconsin	2013
	Yields, Delays, and Mixing in Chemical Evolution Models SDSS-II Collaboration Meeting, Johns Hopkins University	2013
	Decoding Galactic Chemical Evolution with Gas-phase and Stellar Abundances Yale Center for Astronomy & Astrophysics Seminar, Yale University	2013
	The Mass–Metallicity Relation: A Window Into Galaxy Evolution Seminar, National Radio Astronomical Observatory	2013
	The Galaxy Mass–Metallicity Relation: A Sensitive Diagnostic of the Processes That Drive Galaxy Evolution Edward F. Hayes Graduate Research Forum, The Ohio State University	2013
	Characterizing the Distribution of Stars in Chemical Abundance Space APOGEE Collaboration Meeting, The Carnegie Observatories	2013
	Accounting for Stellar Abundance Uncertainties in Principal Component Abundance Analysis APOGEE Collaboration Meeting, Texas Christian University	2012
	Principal Component Abundance Analysis: Finding the Principal Components of Chemical Abundance Space SDSS-III Collaboration Meeting, Vanderbilt University	2011
	Radiation Pressure Feedback in Galaxies Edward F. Hayes Graduate Research Forum, The Ohio State University	2011
POSTERS	Principal Component Abundance Analysis and Chemical Evolution Models The Near-Field Deep-Field Connection, University of California, Irvine	2014
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